CUSTOMER SHOPPING BEHAVIOR ANALYSIS

End-to-End Data Analytics Project

Python

PostgreSQL

Power BI

3,900 Customer Records • 18 Features • 10 SQL Queries

Ibrahim | Luther College | June 2026

AGENDA

01 **Problem Statement**
Business context & objectives

02 **Dataset Overview**
3,900 records · 18 features

03 **Python / EDA**
Cleaning, imputation, feature engineering

04 **SQL Analysis**
10 business queries in pgAdmin

05 **Power BI Dashboard**
Interactive visualizations

06 **Key Findings**
Insights & trends

07 **Recommendations**
Strategic next steps

PROBLEM STATEMENT

The Business Question

"How can the company leverage consumer shopping data to **identify trends, improve customer engagement,** and optimize marketing and product strategies?"

1. **Trends** — Purchasing patterns across demographics & categories
2. **Engagement** — Factors driving repeat purchases & subscriptions
3. **Strategy** — Discount, review, and shipping optimization

3,900 Customer Records	**18** Data Features
10 SQL Queries	**4** Analysis Phases

DATASET OVERVIEW

3,900
Total Records

18
Columns

37
Null Values
(Review Rating only)

Demographics

Age · Gender · Location · Subscription Status

Transaction

Item Purchased · Category · Purchase Amount · Season

Behavior

Discount Applied · Previous Purchases · Frequency · Review Rating · Shipping Type

Payment

Payment Method · Promo Code Used

PYTHON / JUPYTERLAB — Data Preparation & EDA

1 **Data Loading**

Imported CSV with pandas — 3,900 rows, 18 columns confirmed

2 **Null Detection**

df.isnull().sum() → 37 nulls in Review Rating only

3 **Imputation**

Filled with category-level median (not global avg)

4 **Column Standardization**

Lowercase + underscores; renamed purchase_amount_(usd)

5 **Feature: age_group**

pd.qcut() → Young Adult, Adult, Middle-aged, Senior

6 **Feature: freq_days**

Mapped text labels (Weekly=7, Monthly=30...) to numeric days

7 **Redundant Column Drop**

discount_applied == promo_code_used 100% → dropped one

8 **Load to PostgreSQL**

SQLAlchemy + psycopg2 → customer table in customer_behavior DB

BY THE NUMBERS

Descriptive Statistics

44 yrs

Avg Customer Age

Range: 18 – 70

$59.76

Avg Purchase

Range: $20 – $100

3.75★

Avg Review Rating

Range: 2.5 – 5.0

25

Avg Prior Purchases

Range: 1 – 50

2

Gender Values

Male: 2,652 | Female: 1,248

4

Product Categories

Clothing · Accessories · Footwear · Outerwear

SQL ANALYSIS — PostgreSQL via pgAdmin

Pipeline

Python DataFrame

Cleaned & engineered

SQLAlchemy Export

df.to_sql() → customer table

PostgreSQL DB

customer_behavior database

pgAdmin Queries

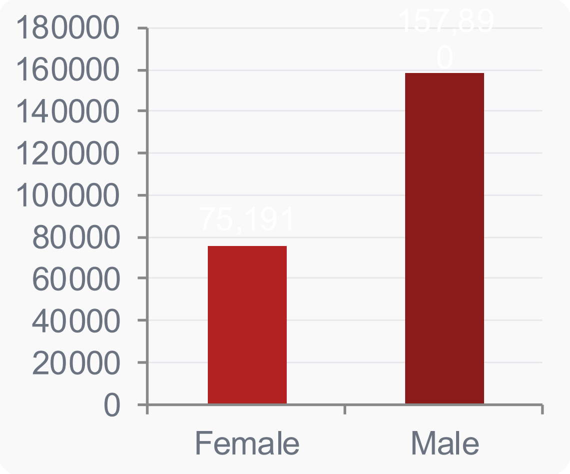
10 business questions answered

10 Queries Executed

- Q1 Revenue by Gender
- Q2 High-Spending Discount Users
- Q3 Top 5 Products by Review Rating
- Q4 Avg Spend by Shipping Type
- Q5 Subscriber vs Non-Subscriber
- Q6 Products with Highest Discount Rate
- Q7 Customer Loyalty Segmentation (CTE)
- Q8 Top 3 per Category (Window Fn)
- Q9 Repeat Buyers & Subscriptions
- Q10 Revenue by Age Group

SQL RESULTS — Revenue, Discounts & Ratings

Q1 Revenue by Gender

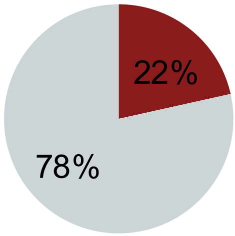


💡 Male 2× Female revenue. Marketing opportunity gap.

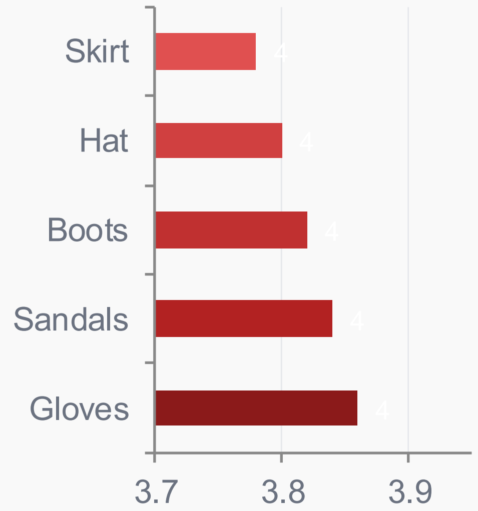
Q2 Discount High-Spenders

839

customers used discounts and STILL spent above the $59.76 average



Q3 Top 5 by Rating



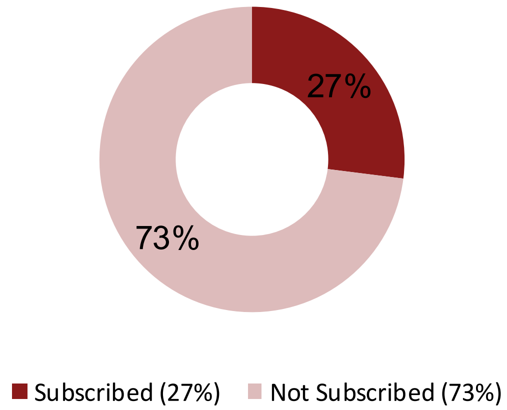
💡 Gloves lead in satisfaction. Feature in campaigns.

SQL RESULTS — Shipping, Subscriptions & Discounts

Q4 Avg Spend by Shipping

💡 Express users spend $2 more. Offer free Express above $75+.

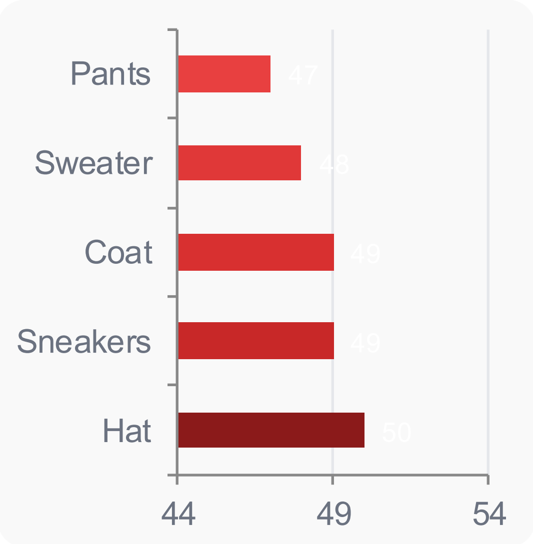
Q5 Subscribers vs Non-Subscribers



Status	Avg $	Total $
Subscribed	$59.49	$62,645
Non-Sub	$59.87	$170,436

💡 Avg spend IDENTICAL — subscribe to capture loyalty, not spend.

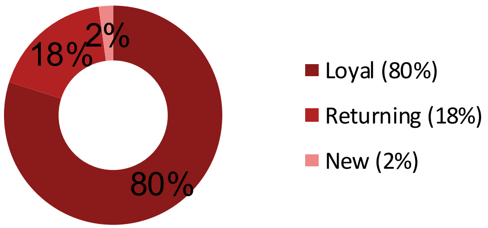
Q6 Top Discount Rate Products



💡 Hat 50% discount rate. Replace blanket discounts with loyalty perks.

SQL RESULTS — Loyalty, Categories & Age Revenue

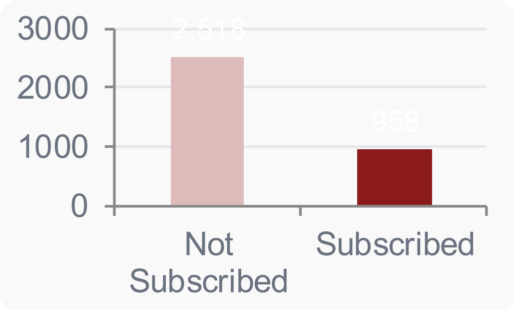
Q7 Customer Loyalty (CTE)



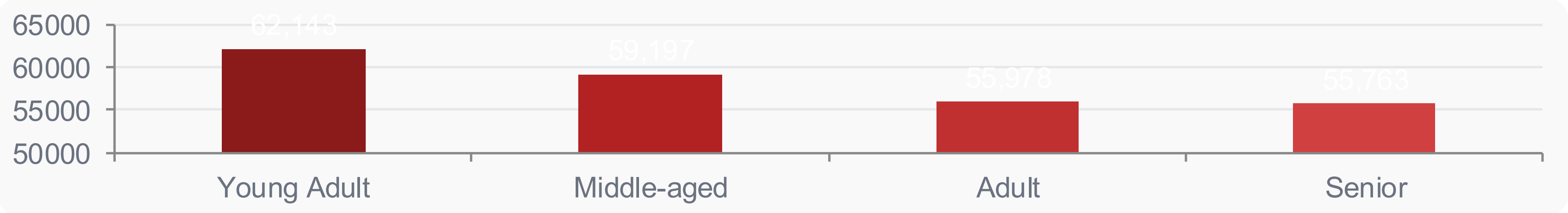
Q8 Top 3 per Category

Category	#1	#2	#3
Clothing	Blouse	Pants	Shirt
Accessories	Jewelry	Sunglasses	Belt
Footwear	Sandals	Shoes	Sneakers
Outerwear	Jacket	Coat	—

Q9 Repeat Buyers & Subs



Q10 Revenue by Age Group




CUSTOMER BEHAVIOR DASHBOARD
3.9K
Number of customers
$59.76
Average Purchase Amount
3.75
Average View Rating
Subscription Statu
No
Yes
Gender
Female
Male
Category
Accessories
Clothing
Footwear
Outerwear
Shipping Type
2-Day Shipping
Express
Free Shipping
Next Day Air
Standard
Store Pickup
% of Customers by Subscription Status
Yes 1.05K
No 2.85K
Revenue by Catagory
100K
50K
0K
Clothing
Accessories
Footwear
Outerwear
Sales by Catagory
2,000
1,500
1,000
500
0
Clothing
Accessories
Footwear
Outerwear
Revenue by Age Group
young Adult
Middle-aged
Adult
Senior
0K
20K
40K
60K
Sales by Age Group
young Adult
Middle-aged
Senior
Adult
0
500
1,000

POWER BI DASHBOARD

Interactive Visualization

3 KPI Cards

Customers · Avg Purchase · Avg Rating

Donut Chart

73% non-subscribed vs 27% subscribed

Revenue by Category

Clothing leads at ~$100K total

Sales by Category

Clothing: 1,750 transactions — dominant

Revenue by Age

Young Adults slightly ahead of all groups

4 Interactive Slicers

Gender · Category · Subscription · Shipping

Dashboard includes real-time filtering — all charts update on slicer selection (Gender, Category, Subscription Status, Shipping Type)

KEY FINDINGS

Revenue Gap

Male customers = 68% of revenue ($157,890 vs $75,191 female). A significant demographic imbalance.

80% Loyal

3,116 of 3,900 customers have 11+ purchases. Retention is exceptional — focus on maximizing their value.

Discounts Work

839 discount users still spent ABOVE average. Discounts attract high-intent, not just budget buyers.

Express = Higher $

Express shipping customers spend $60.48 vs $58.46 standard. Upsell shipping to boost basket size.

Top-Rated Products

Gloves (3.86), Sandals (3.84), Boots (3.82) — highest satisfaction. Feature these in campaigns.

73% Unsubscribed

2,847 non-subscribers including 2,518 repeat buyers. Largest single conversion opportunity.

Clothing Dominates

$100K revenue, 1,750 transactions — double the next category (Accessories). Core growth driver.

Equal Age Revenue

All 4 age groups within 10% of each other. Broad cross-generational appeal — no narrow targeting.

STRATEGIC RECOMMENDATIONS

Business Next Steps

1 Subscription Campaign

HIGHEST PRIORITY

Target 2,518 repeat, non-subscribed buyers with first-month-free offers. Set 20% conversion goal within 6 months.

2 Gender Marketing

HIGH PRIORITY

Audit female ad spend. Design Accessories/Footwear campaigns targeting female segments. A/B test personalized messaging.

3 Product Spotlight

MEDIUM PRIORITY

Feature top-RATED items (Gloves, Sandals, Boots) on homepage. Investigate quality gap in top-selling items.

4 Smart Discount Policy

MEDIUM PRIORITY

Switch from blanket discounts to tiered incentives tied to basket size. Test loyalty points as discount alternative.

5 Shipping Optimization

QUICK WIN

Offer free Express above $75 spend threshold. Drives $2+ higher basket size per transaction.

6 Loyalty Tier Program

LONG TERM

Introduce Silver/Gold/Platinum tiers. Nudge 701 Returning customers toward Loyal status with bonus incentive.

TOOLS & TECHNOLOGY STACK

Python / JupyterLab

- pandas — data loading & cleaning
- pd.qcut() — feature engineering
- fillna with lambda — null imputation
- SQLAlchemy / psycopg2 — DB export

PostgreSQL / pgAdmin

- 10 analytical SQL queries
- GROUP BY / SUM / AVG / COUNT
- Subquery with scalar SELECT
- CTE (WITH clause)
- ROW_NUMBER() window function

Power BI

- KPI cards — 3 metrics
- Bar & donut charts
- Revenue & sales by age/category
- 4 interactive slicers (real-time)

CONCLUSION

End-to-End Analytics, Real Business Impact

- ✓ Cleaned & engineered 3,900 records across 18 features using Python
- ✓ Executed 10 SQL queries (aggregations, subqueries, CTEs, window functions)
- ✓ Built interactive Power BI dashboard with 4 real-time slicers
- ✓ Identified 2,518 repeat buyers not subscribed — highest ROI target
- ✓ Proved discounts attract high-intent buyers, not budget shoppers

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Thank You | Questions?